

Activity 04: Relational Schema Normalization

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1. Objectives

- Identify functional dependencies within an unnormalized data set.
- Convert a flat relation into Third Normal Form (3NF).
- Verify referential integrity constraints across newly decomposed relations.

Key Constraint: Foreign keys must explicitly reflect cascading delete rules defined in Section 3.2.

2. Relational Decompositions & Normalization State

The table below summarizes the normalization progression for each decomposed entity:

Table Name	Primary Key	Normal Form	Status
tbl_students	student_id	3NF	VERIFIED
tbl_courses	course_id	3NF	VERIFIED
tbl_enrollments	enrollment_id	3NF	VERIFIED
tbl_instructors	instructor_id	3NF	PENDING

3. Verification & Integrity Summary

All transitively dependent attributes (department_name , office_location) have been decoupled from the primary entity key into lookup relations, thereby eliminating update, insertion, and deletion anomalies.